

Abstract

To assist the Clearing Of Multi-use Assembly Public/Private spaces (COMAP) in planning the best route for responders to sweep the floor of buildings during an emergency, a comprehensive optimization for sweeping routes is accomplished by our projecting model, aiming to minimize total room-clean time while prioritizing occupants' safety and warranting responders' efficiency.

For initial consideration, fire emergency has been chosen as the target emergency, occurring in office buildings, restaurants, and school labs. Later on, the model is extended to plan sweep routes for responders in other emergencies involving dispersible and harmful materials, like gas leaks.

In the main model, several distinct factors have been taken into consideration: fire propagation, smoke propagation, occupants' movement dynamics, and the importance of the room. Among which the fire propagation and smoke propagation are influenced by air flow rate and direction, types of combustible materials, and building structure. Occupants' movements are blocked by fire spread, and their velocities vary with their ages as well as occupant flow density. The importance of a room is evaluated through its risk level and the probability of someone with limited mobility within the room. These factors are studied through Cellular Automaton (CA), in which the potential fields of static building structure, fire spread, smoke spread, and occupant flow density are established. Potential fields show the tendency of movement in each cell, making it a favorable choice for simulating movement of different subjects.

The major optimization process is achieved through the application of a modified version of the MILP, combining CP-SAT and Cellular Automaton. The influences of the four main factors on responders' path choice are manifested through a reward function and modification of responders' speed and vision range. MILP is conducted to gain the optimal path and number of responders that ensure scanning every cell while minimizing sweeping time.

At first, our model is utilized to predict the best route for two firefighters in an office building. Then, it is extended to various constructions with different floors and structures, including a restaurant and a school lab. We also applied the model to another emergency—a gas leak—in which we considered occupants' delayed awareness of the emergency.

We tested our model by using the Fourier Amplitude Sensitivity Test. Results show that our model is able to assess the best plan for different numbers of responders to expend the shortest time to search every corner inside of the building. It is also proved that not only in fire emergencies but also in gas leaks, our model is always capable of giving out the optimal route.

Key Words: Emergency Evacuation, Cellular Automaton, MILP, Potential Field, Fourier Amplitude Sensitivity Test.

Contents

1	Introduction	1
1.1	Background	1
1.2	Restatement of the Problem	1
1.3	Our Work	2
2	Assumptions and Justifications	2
3	Notations	3
4	Impact Factors	4
5	Methodology and Model Setup	5
5.1	Framework of the Model	5
5.2	Mixed-integer Linear Programming	5
5.3	Cellular Automaton	6
5.4	Preset Scenario	7
6	Model Implementation	8
6.1	Application of the Cellular Automaton	8
6.1.1	Fire Propagation	8
6.1.2	Smoke Propagation	9
6.1.3	Setup of the Static Potential Field	11
6.1.4	Potential Field for the Crowd Density	11
6.1.5	Occupant Movement Dynamics	12
6.1.6	Door Capacity Constraint	13
6.2	Integration of CP-SAT and Cellular Automaton	14
6.2.1	Architecture of the Model	14
6.2.2	The Coupling Pipeline	14
6.3	Application of the CP-SAT Method	15
6.3.1	Data Preprocessing	15
6.3.2	Constraints	16
6.3.3	Objective Function	17
6.3.4	Real-life Applications: The Different Rooms	18
6.4	Model Results	18
7	Sensitivity Analysis	19
7.1	Theoretical Basis of FAST	20
7.2	Application in our Model	20
8	Strengths and weaknesses	20
9	Conclusion	21

1 Introduction

1.1 Background

Seconds count during emergencies like fires and gas leaks. A well-coordinated evacuation of the area is the thin line separating safety and danger. A hurried ‘clear’ of a smoky office floor could beat the clock but tragically leave an unconscious person behind; on the other paw, an extremely meticulous sweep of the building could prove disastrous due to the rapidly advancing fire. Such conflicting forces of haste and vigilance clearly point out the deep-seated requirement of an ‘optimized’ approach.

Therefore, there’s an increased need to form an optimal mathematical approach that could address these sweep strategies effectively. This mathematical approach could ensure that the time taken to clear the building is minimized and the probability of a complete sweep is maximized, directly leading to the saving of lives.



1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- **Build the Model.** Create a mathematical model to find the optimal sweep path and the shortest completion time.
- **Validate the Model.** Examining the model’s performance on the single-story office building with six rooms and two responders to verify its logic and accuracy.
- **Report the Findings.** Based on the validated model, apply it to at least two additional scenarios with different structures of the buildings.

- **Analyze Key Factors.** Conduct a sensitivity analysis to quantify the impact of key variables on the optimal sweep strategy and completion time.
- **Report the Findings.** Write a concise letter to the emergency committee with our model's recommendations.

1.3 Our Work

Our major innovation consists of a decision-support system with a two-stage structure that intertwines the simulation of the environment based on the CA approach with the optimal path-routing calculation of the MILP method. This system improves its performance over time during the iterative development process and thus proves its scalability even in highly complex emergencies.

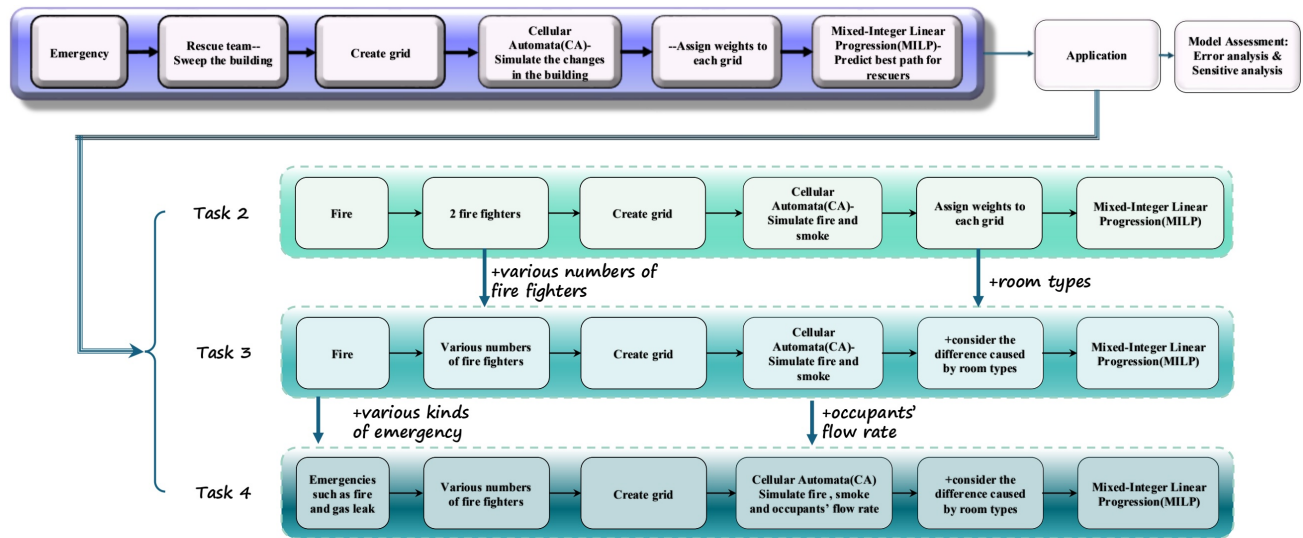


Figure 1: Flowchart of the Model

2 Assumptions and Justifications

To render the problem tractable and focus on its core mechanisms, we establish the following foundational assumptions. Each assumption is accompanied by a justification to ensure the effectiveness of our model.

- **Assumption 1: The building's topology structure remains invariant, regardless of the fire's intensity.**
Justification: Because fire-driven structural damage is highly random, we assume a static layout so we can focus on optimizing responders' search strategy rather than trying to track uncertain building changes.
- **Assumption 2: All else being equal, responders will prioritize areas with greater fire intensity.**

Justification: We make this simplification to capture the core driver of responder behavior—focusing on the most severe fire to prevent spread—and to turn complex real-world judgments into a single quantifiable rule that keeps the model computationally tractable.

- **Assumption 3: We have two types of firefighters: 'Search Firefighters' and 'Rescue Firefighters'. However, in our case, the firefighters are only responsible for the search task, not the rescue task.**

Justification: This distinction lets us focus on the core task of the sweep—optimizing paths and verifying information—without the added complexity of rescue logistics, so the model can better solve the primary optimization problem.

- **Assumption 4: We assume that escape routes are clear of obstructions.**

Justification: We exclude blocked routes to avoid complex stampede and obstacle dynamics, keeping the environment stable and predictable so we can focus on designing an effective sweep strategy.

3 Notations

Table 1: Key Notation Used in the Model

Symbol	Description	Unit	Symbol	Description	Unit
i, j	Row and column indices of the cellular grid	–	m	Flattened index of a grid cell (node)	–
t	Discrete time step index	s (step)	N	Number of firefighters (responders)	–
M	Number of passable grid cells (nodes)	–	T	Number of time steps in the time horizon	–
$F(i, j, t)$	Fire state of cell (i, j) at time t	–	$S(i, j, t)$	Smoke state of cell (i, j) at time t	–
$P_{\text{fire-fix}}$	Fixed component of fire spread probability	–	$P_{\text{fire-evolv}}$	Dynamic component of fire spread probability	–
P_{fire}	Total probability of ignition for a cell	–	P_{smoke}	Probability of smoke propagation to a cell	–
q_{rad}	Incident radiative heat flux	kW/m ²	q_{crit}	Critical heat flux of combustible material	kW/m ²
m_c	Fuel mass contained in one combustible cell	kg	r_c	Mass loss rate of fuel	kg/(m ² s)

Table 1 (continued)

Symbol	Description	Unit	Symbol	Description	Unit
A	Area of a single grid cell (0.4×0.4)	m^2	t_c	Burnout time of a com- bustible cell	s
$\Phi(i, j)$	Static potential (distance) – from cell (i, j) to the exits	–	$\Phi_{\text{norm}}(i, j)$	Normalized static potential – field	–
$\rho_{\text{raw}}(i, j)$	Raw number of occupants at cell (i, j)	persons	$\rho(i, j)$	Gaussian-smoothed occu- pant density at (i, j)	persons/ m^2
$\rho_{\text{norm}}(i, j)$	Normalized occupant density – field	–	D	Local crowd density factor – used in speed reduction	–
v_0	Base walking speed of a re- sponder in free flow	m/s	v	Effective responder speed under congestion	m/s
$s(t, m)$	Local reachable radius from node m at time t	cells	$c(t, m)$	Visibility radius from node m at time t	cells
P_{ij}	Transition probability for an – occupant to move to neighbor (i, j)	–	E	Set of exit cells in the grid –	–
V	Set of all nodes	–	M^*	Set of high-priority nodes – that must be covered	–
$\text{reachable}(t, p)$	Set of nodes reachable from – node p at time t	–	$\text{vision}(t, p)$	Set of nodes visible from – node p at time t	–
$\text{watchers}(t, m)$	Nodes from which cell m is – visible at time t	–	$r(t, m)$	Time-dependent reward as- signed to node m	–
$r'_{t,m}$	Differenced reward used in – the objective function	–	$w_{n,t,m}$	Binary: firefighter n at node – m at time t	–
$y_{t,m}$	Binary: at least one fire- – fighter at node m at time t	–	$v_{t,m}$	Binary: node m is visible at – time t	–
$b_{t,m}$	Binary: node m has been – searched up to time t	–			

4 Impact Factors

In predicting the optimal rescue route, our model integrates three primary categories of influencing factors: dynamic disaster elements, responder characteristics, and static environmental features. As illustrated, these encompass the fire's spread and smoke (simulated by our CA module), the responders' number, speed, and field of view (which act as key constraints in the MILP model), and the room's

function and fuel load, which determine search priorities. Together, these factors define the risk and value of each location over time, thereby guiding the generation of the optimal path.

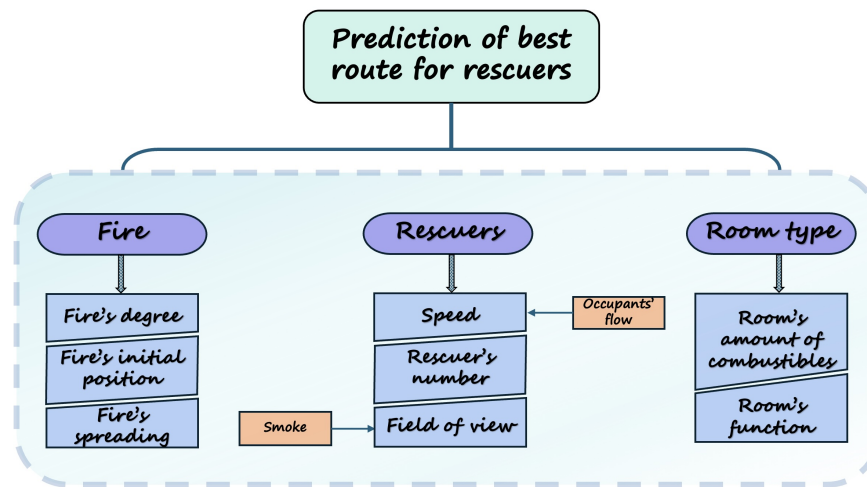


Figure 2: Impact Factors

5 Methodology and Model Setup

5.1 Framework of the Model

Our model is a two-stage pipeline that couples environmental simulation with path optimization. First, a Cellular Automata (CA) simulates the disaster's evolution, generating time-series data on fire, smoke, and occupants. This data then informs a Mixed-Integer Linear Programming (MILP) model, which solves for the optimal rescue paths to maximize search efficiency under dynamic constraints.

5.2 Mixed-integer Linear Programming

- **Basic Information.** Mixed-integer linear programming (MILP) is an optimization technique used to find the best solution to problems with a linear objective function and constraints, where some variables must be integers while others can be continuous. It's a powerful tool for combinatorial optimization problems, which are common in real-world scenarios, and can be used for things like production planning, logistics, and smart grid energy management.
- **The Reason We Apply.** We use Mixed-Integer Linear Programming (MILP) because it naturally captures discrete responder–room assignments and continuous timing, allowing us to optimally schedule the sweep and minimize total search time.
- **Evaluating the Effectiveness and Robustness.**
 - ◆ **Effectiveness.** We use Mixed-Integer Linear Programming (MILP) because it naturally captures discrete responder–room assignments and continuous timing, allowing us to optimally schedule the sweep and minimize total search time.

- ◆ **Robustness.** The same formulation can handle multi-story layouts, different responder numbers, and room priorities (e.g., a daycare) without changing the algorithm.

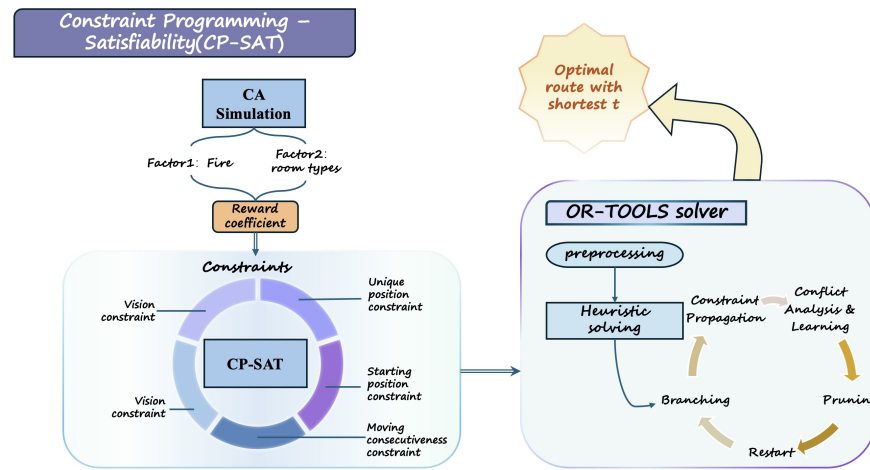


Figure 3: Flowchart of CP-SAT

5.3 Cellular Automaton

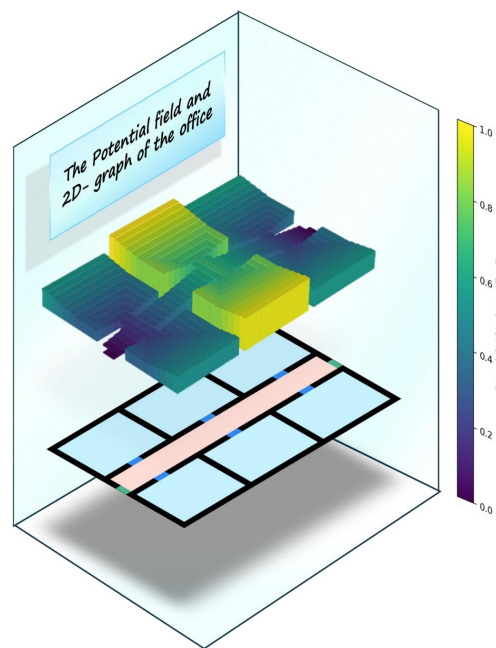


Figure 4: The Potential Field and 2D-graph of the Office

- **Basic Information.** Our CP-SAT MILP model stays the same while adapting to different layouts, responder counts, and room priorities. A cellular automaton (CA) uses simple local rules on a grid to simulate fire spread and other spatial dynamics.

- **The Reason We Apply.** By adjusting only the grid layout and a few fire-spread parameters, it can handle different buildings and fire speeds while still finding a feasible rescue plan.
- **Evaluating the Effectiveness and Robustness.**
 - ◆ **Effectiveness.** Our CA model continuously updates a danger map of the fire, and the pathfinding model uses it to adapt rescue routes in real time.
 - ◆ **Robustness.** By adjusting only the grid layout and a few fire-spread parameters, it can handle different buildings and fire speeds while still finding a feasible rescue plan.

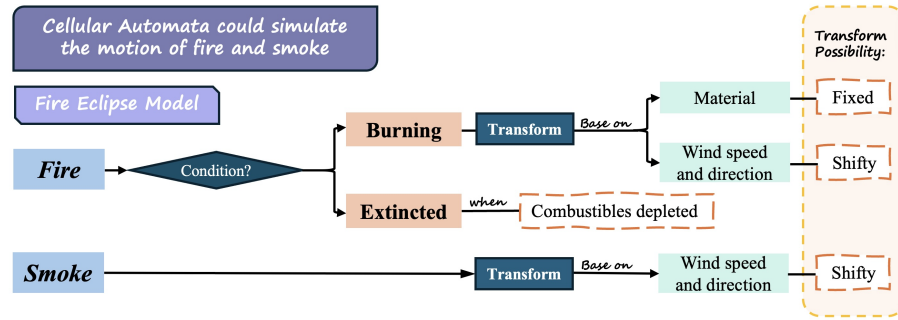


Figure 5: Flowchart of Cellular Automaton

5.4 Preset Scenario

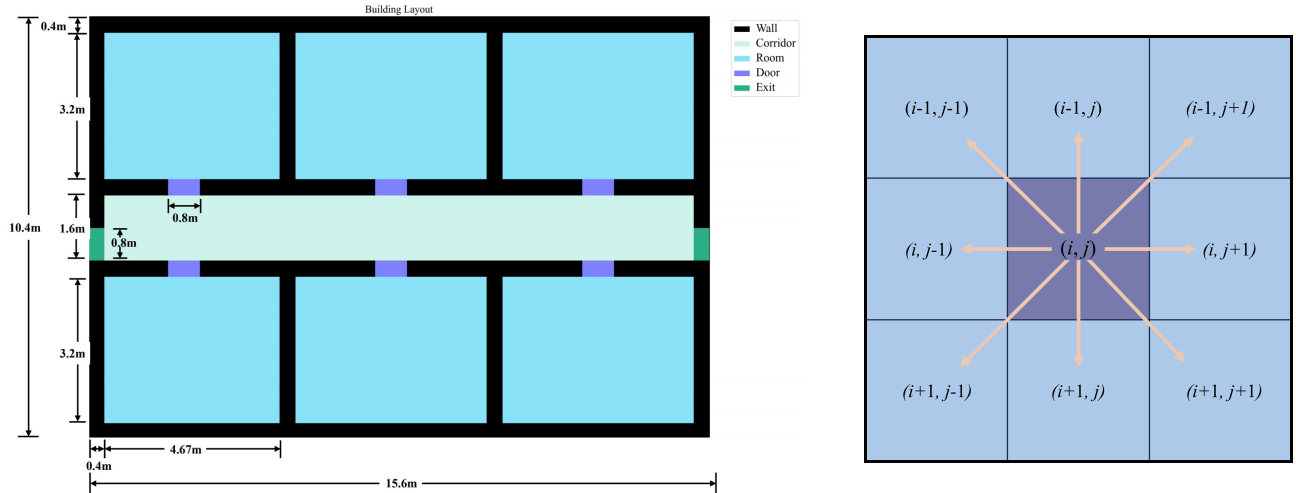


Figure 6: Building Layout

Figure 7: Movement Logic

We illustrate our model on a residential building whose floor plan is discretized into a 26×39 grid of $0.4 \text{ m} \times 0.4 \text{ m}$ cells (Figure (6)). This cellular representation is layout-agnostic and can be applied to any floor plan; cell locations are denoted by index pairs (i, j) .

Moore neighborhood is employed to define local interactions between cells. For a central cell (i, j) , the Moore neighborhood comprises the eight adjacent cells, including both orthogonal and diagonal neighbors, that are influenced by the central cell, as shown in Figure (7)

6 Model Implementation

6.1 Application of the Cellular Automaton

6.1.1 Fire Propagation

In the fire field, each cell in the cellular grid has a discrete state with the following meanings:

$$F(i, j) = \begin{cases} 0, & \text{empty.} \\ 1, & \text{unburned.} \\ [2, \infty), & \text{burning.} \end{cases} \quad (1)$$

where empty means the cell is a wall and thus non-combustible; Unburned means the cell is not yet burning but is combustible; Values greater than 2 mean the cell is burning, and larger integer values indicate higher fire intensity.

Propagation Probability. Initially, the probability is then broken down to two parts that contribute to fire spread: the fixed probability and the evolving probability:

- **Fixed probability.** It is a fixed ignition probability determined by the combustible material in that cell. We only model thermal radiation as the heat transfer mode because it dominates remote ignition in large fires and keeps the model simple. A cell ignites when $q_{\text{rad}} > q_{\text{crit}}$, where q_{crit} is the minimum heat flux required to ignite the material and q_{rad} is the incident heat flux from the fire; a smaller q_{crit} means easier ignition. The value of q_{crit} depends on the fuel type; in an office we consider chairs, wooden desks, bookshelves, and carpets, etc.

Table 2: Combustible Material Properties

Combustible Material	Raw Material	Critical Heat Flux $q_{\text{crit}}/(kW \cdot m^{-2})$	Mass Loss Rate $r_c/(g \cdot m^2 \cdot s)$	Mass per Cell m/kg
Wooden desk	European Spruce	22	13	5
Bookshelf	European Spruce	12.5	13	5
Chair	Flexible polyurethane foam	13	10	0.33
Carpet	Nylon	20	15.1	0.08

In each office, we randomly generate a 1×1 chair, a 1×2 bookshelf, and a 1×2 desk. Additionally, a 3×3 carpet is generated in three of the rooms, and we have

$$P_{\text{fire-fix}} = \begin{cases} 0, & \text{if } (i, j) \text{ has no combustible object.} \\ \frac{1}{2 + 2e^{q_{\text{incident}} - q_{\text{crit}}}}, & \text{if } (i, j) \text{ has combustible object } (q_{\text{incident}} = 25kW/m^2s) \end{cases} \quad (2)$$

➤ **Evolving P .** It is based on the Fire Propagation Ellipse Model, which is

$$P_{\text{fire-evolv}} = \|\text{segment}\| \quad (3)$$

Additionally, if a cell is evaluated multiple times, the maximum value is assigned to it. Apply the normalization by the maximum value. Normalizing $P_{\text{fire-fix}}$ as

$$P_{\text{fire-fix}} = 0.5k_1 \cdot \frac{P_{\text{raw}}}{P_{\text{max}}} \quad (4)$$

Overall, it follows

$$P_{\text{fire}} = P_{\text{fire-fix}} + P_{\text{fire-evolv}} \quad (5)$$

On top of that, the fire is extinguished as the fuel burns out. Let t_c denoted by the total time. Thus, the propagation process of fire F satisfies the following rule:

$$F(i, j) = \begin{cases} 0, & \text{if empty.} \\ 1, & \text{if } F(i, j)_{t-1} = 1 \text{ and random } P_{\text{fire}} < 0.5. \\ [2, \infty), & \text{if } F(i, j)_{t-1} = 1 \text{ and random } P_{\text{fire}} \geq 0.5. \\ F(i, j)_{t-1} + 1, & \text{if } F(i, j)_{t-1} \geq 2 \text{ and } t \leq \frac{t_c}{2}. \\ F(i, j)_{t-1} - 1, & \text{if } F(i, j)_{t-1} \geq 2 \text{ and } t \geq \frac{t_c}{2}. \end{cases} \quad (6)$$

with

$$t_c = \frac{m_c}{r_c \cdot A} \quad (7)$$

where A is the cell area that can be calculated as $0.4 \cdot 0.4 = 1.6$, m_c is the mass of the cells, and r_c is the mass loss rate.

6.1.2 Smoke Propagation

➤ Similar to the fire propagation, consider three states of the smoke S ,

$$S(i, j) = \begin{cases} 0, & \text{empty.} \\ 1, & \text{without smoke.} \\ [2, \infty), & \text{with smoke.} \end{cases} \quad (8)$$

➤ **Propagation Probability.** The probability of smoke propagation also satisfies the Fire Propagation Ellipse Model, which is

$$P_{\text{smoke}} = \frac{k_2 \cdot \|\text{segment}\|}{\max(\|\text{segment}\|)}. \quad (9)$$

where

$$\|\text{segment}\| = k_2 \cdot \|\vec{v}\| \quad (10)$$

Thus, the propagation process of smoke S satisfies the following rule:

$$S(i, j) = \begin{cases} 0, & \text{if } S_m(i, j)_{t-1} = 0. \\ 1, & \text{if } S_m(i, j)_{t-1} = 1 \text{ and } P_{\text{smoke}} < 0.5. \\ 2, & \text{if } S_m(i, j)_{t-1} = 1 \text{ and } P_{\text{smoke}} \geq 0.5. \\ S_m(i, j)_{t-1} + 1, & \text{if } 2 \leq S_m(i, j) \leq 10. \end{cases} \quad (11)$$

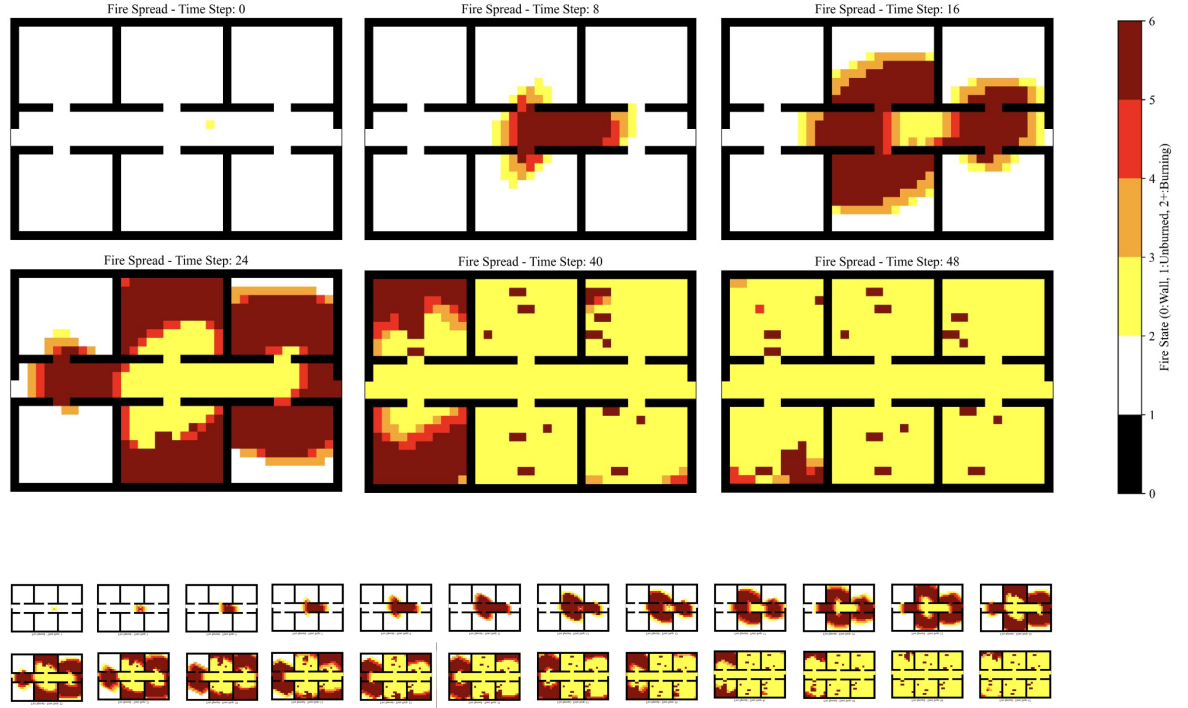
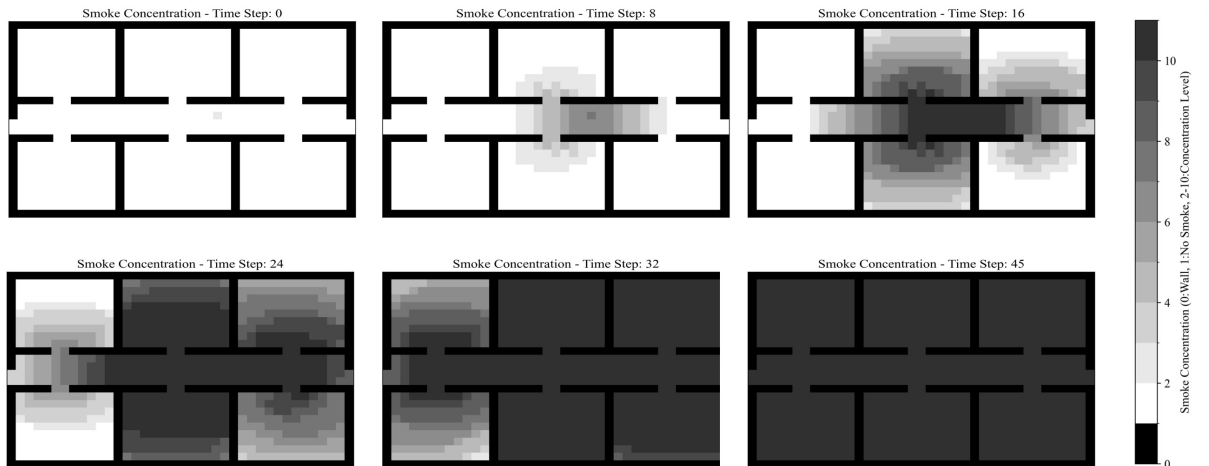


Figure 8: Fire Propagation Dynamics



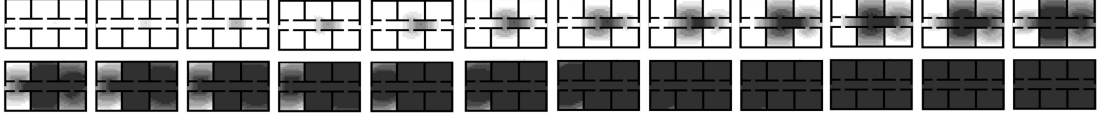


Figure 9: Smoke Propagation Dynamics

6.1.3 Setup of the Static Potential Field

Our static potential field is essentially a distance map generated by a Breadth-First Search (BFS) algorithm. The closer a cell is to an exit, the lower its potential value (i.e., distance), creating a clear gradient that guides people along the optimal path.

- Initialize the Potential Field: $\Phi(i, j) = \infty$
- Let E be the set of all exit cells. For each cell $m \in E$

$$i_e \in 11, 12 \text{ and } j_e \in 0, 38, \quad \Phi(i, j) = 0 \quad (12)$$

- From the exit cells, the algorithm expands to their immediate neighborhood.

The model defines cell adjacency using the Moore neighborhood, which means a cell is considered a neighbor if it is connected orthogonally or diagonally. Thus, each cell has up to eight neighbors.

i.e., for each position (i_n, j_n) :

$$(i, j) \in \{(i_0, j_0) + \Delta | \Delta \in \{(-1, -1), (-1, 0), (-1, 1), (0, -1), (0, 1), (1, -1), (1, 0), (1, 1)\}\}. \quad (13)$$

If the grid is traversable (i.e., not a wall), and $\Phi(i_0, j_0) < \Phi(i_n, j_n)$, then

$$\Phi(i_n, j_n) = \Phi(i_0, j_0) + 1. \quad (14)$$

Thus, the grid is processed in layers based on the shortest distance from each cell to an exit, a procedure executed by a Breadth-First Search (BFS). Subsequently, these raw potential values undergo a max-min normalization,

$$\Phi_{\text{norm}}(i, j) = \frac{\Phi(i, j) - \min(\Phi)}{\max(\Phi) - \min(\Phi)}. \quad (15)$$

which transforms the absolute distances into a normalized potential field that represents relative proximity to safety and maintains a uniform gradient for pathfinding.

The potential of wall

$$\Phi_{\text{norm}}(i, j) = \infty. \quad (16)$$

6.1.4 Potential Field for the Crowd Density

- $\rho_{\text{raw}}(i, j)$ denote the number of occupants at a given cell m .
- **Gaussian Filtering.** To obtain a smooth occupant density field, we perform a convolution operation on the discrete occupant counts with a Gaussian kernel:

$$P(i, j) = \sum_{m=-k}^k \sum_{n=-k}^k G(m, n) \cdot \rho_{\text{raw}}(i + m, j + n). \quad (17)$$

where $G(m, n)$ is the Two-dimensional Gaussian kernel function

$$G(m, n) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{m^2 + n^2}{2\sigma^2}\right). \quad (18)$$

The standard deviation for the Gaussian kernel is set to $\sigma = 1.5$, and the kernel size is defined as 7×7 , where ($k = 3$).

This process effectively simulates how a responder would perceive the overall density of their immediate surroundings, ensuring that our crowd-induced potential field is based on a smooth and realistic representation of congestion.

➤ **Normalization by the maximum value.**

$$\rho_{\text{norm}}(i, j) = \frac{\rho_{\text{smooth}}(i, j)}{\max(\rho_{\text{smooth}})} \quad (19)$$

The density of the wall

$$\rho_{\text{norm}} = \infty \quad (20)$$

6.1.5 Occupant Movement Dynamics

- **Initialization.** Each room is randomly populated with a number of occupants drawn from a uniform integer distribution between 2 and 4. Each occupant is modeled as an agent occupying a single grid cell. Given the six rooms in the scenario, this results in a total population of 12 to 24 agents.
- **Initial velocity.** $v_0 = 1.2m \cdot s^{-1}$.
- **Logic of Occupant Movement.** Each pedestrian can move to one unoccupied neighbor cell (i, j) (or stay at the present cell) at each discrete time step $t \rightarrow t + 1$ according to a certain transition probability P_{ij} ,

$$P_{ij} = \frac{N}{S_{ij} F_{ij}^2} \eta_{ij} \varepsilon_{ij}. \quad (21)$$

where

$$\eta_{ij} = \begin{cases} 0, & \text{cell is empty.} \\ 1, & \text{cell is occupied by wall or obstacle.} \end{cases} \quad (22)$$

$$\varepsilon_{ij} = \begin{cases} 0, & \text{cell is empty.} \\ 1, & \text{cell is occupied by an occupant.} \end{cases} \quad (23)$$

N is a normalization factor for ensuring $\sum_{i,j} P_{ij} = 1$.

- **Velocity of Occupant Movement.** Occupants are categorized into four types: the elderly $[60, \infty)$, young people $[18, 60)$, children $[6, 18)$, and parents with young children $[0, 6)$ that are treated as the elderly. The base speed for young people is set to $3.64m/s$, whereas the elderly, children, and parents with young children share a base speed of $1.212m/s$. This distinction is also reflected in the simulation's update frequency: young people are updated three times per time step, while all other occupant types are updated once per time step.

- F_{ij} here stands for the fire field, where the previous one was applied in the state of fire. F_{ij} can be expressed as

$$F_{ij} \begin{cases} \geq 0, & \text{if } \tau \leq \gamma. \\ = 0, & \text{if } \tau \geq \gamma. \end{cases} \quad (24)$$

where The parameter τ shows the distance from cell (i, j) to the border of the fire, γ is 6 in our case.

For the parameter α in F_{ij} : when F_{ij} and the distance from the cell (i, j) to the exit is closer than σ (in this paper, $\sigma = 6$), α is taken as 0.5; else, it is taken as 1. The reason is that when pedestrians are near to the exit, even though there is fire around them, the instinct of survival makes the influence of the fire smaller; thus, the probability that pedestrians run toward the exit is larger.

- **Velocity Variation Due to Crowd Density.** Recognizing that crowd congestion is a critical bottleneck in emergencies—making the shortest path not always the fastest—we model responder velocity (v) as a function of crowd density (D). This compels our model to find a truly time-optimal path, rather than just the geometrically shortest one.

$$v = v_0 e^{-2D^2}. \quad (25)$$

where v_0 is a fixed value $5m/s$.

Occupant Evacuation– Time Step at 0, 5, 10, 14, 18, 23

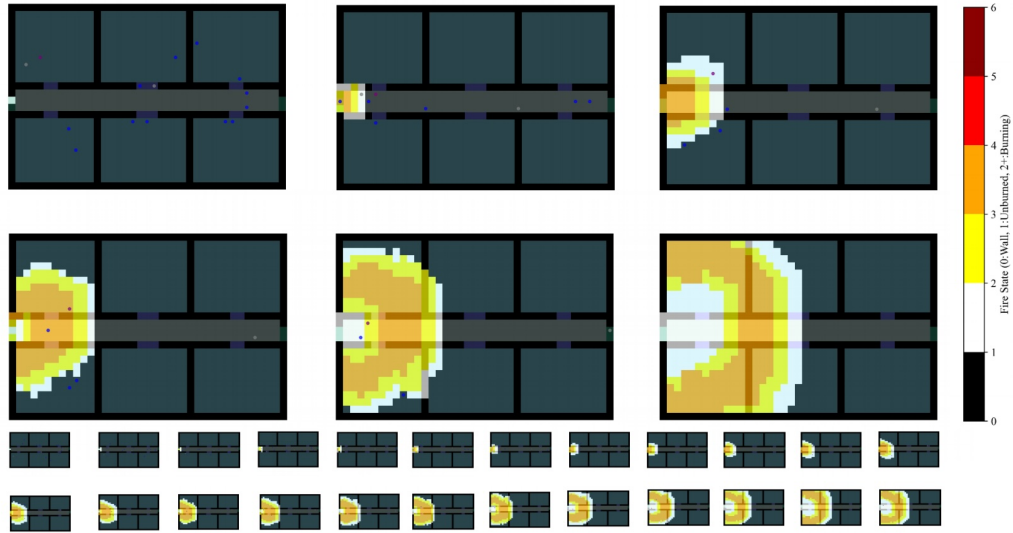


Figure 10: Occupant Evacuation

6.1.6 Door Capacity Constraint

- The maximum number of occupants that can pass through a single doorway per time step

$$C_{\max} = 2. \quad (26)$$

- When $C_{\text{current}} < C_{\max}$, it can pass through.

6.2 Integration of CP-SAT and Cellular Automaton

6.2.1 Architecture of the Model

Conceptually, the model is structured as a two-layer coupled system.

- **Underlying Layer.** This component uses a Cellular Automaton (CA) to update dynamic fire, smoke, and occupant density fields, which are then combined with static exit-distance and congestion data to construct a time-varying graph.
- **Application Layer.** The CP-SAT model discretizes the entire building into a time-expanded graph, which is defined as follows:
 - ◆ **Spatial Vertices.** Each passable cell m .
 - ◆ **Time Layers.** Each time step t from 0 to $T - 1$.
 - ◆ **Decisions on the Graph.** These variables determine which cell each responder n occupies at each time layer, as well as which cells they observe or search at any given moment.

6.2.2 The Coupling Pipeline

Offline CA Module Simulation.

- **Inputs.** Building layout, initial fire source location, distribution of combustible materials (e.g., desks, chairs, bookshelves, carpets), and initial occupant positions.
- **Update Logic.** The simulation evolves based on the rules defined in Subsection 6.1:
 - ◆ **Fire Field $F(i, j, t)$.** Updated using a hybrid model of fixed and evolutionary probabilities, with the burnout time governed by mass loss.
 - ◆ **Smoke Field $S(i, j, t)$.** Propagates based on the propagation ellipse model.
 - ◆ **Occupant Density Field $P(i, j, t)$.** First simulating individual movements based on a potential field and then applying Gaussian smoothing to obtain a continuous density field.
- **Outputs.** The simulation results are written to the CSV file.

Intermediate Processing Layer. Converting CA Outputs into CP-SAT Graph Constraints.

- **The local reachable radius $s(t, m)$** is calculated from the occupant density $P(i, j, t)$ to model the speed reduction caused by congestion. The formula is given by:

$$s(t, m) = \left\lfloor \frac{25}{6} e^{-2P(t, m)} + \frac{1}{2} \right\rfloor. \quad (27)$$

As the density P increases, the exponential term decreases, which in turn reduces the radius s . Consequently, the reachable range of the cell is contracted at that time step.

- **The visibility radius $c(t, m)$** is calculated based on the smoke state $S(i, j, t)$:

$$c(t, m) = \lfloor 15 - S(i, j, t) \rfloor. \quad (28)$$

The thicker the smoke, the shorter the visibility distance.

Optimal Responder Paths with the CP-SAT Module.

- A CP-SAT model is formulated with the following parameters: the number of responders N , the time horizon T , the number of cells M , and the pre-computed values for reachable, vision, and r .
- Upon solving, the model outputs the path for each responder, along with the optimal objective value, which represents the total search value.

6.3 Application of the CP-SAT Method

6.3.1 Data Preprocessing

Model Formulation. The building floor is discretized into a set of nodes, which can represent grid cells or rooms. The key components are defined as follows:

- **Set of Nodes:** The set of nodes is denoted by $V = 0, \dots, M - 1$.
- **Time Discretization:** The time horizon is discretized into T steps, indexed by $t = 0, \dots, T - 1$.
- **Firefighters:** There are N firefighters, indexed by $n = 0, \dots, N - 1$.

CA Fire Simulation Inputs. In the CP-SAT module, we map the two-dimensional index (i, j) to a one-dimensional cell index m , and treat the CA fire field as a time-dependent reward field $r(t, m)$. Concretely, for each time step t and each grid cell m , we set

$$r(t, m) = F(i, j, t), \quad (29)$$

where m is the flattened index corresponding to cell (i, j) . A larger value of $r(t, m)$ indicates that visiting cell m at time t is more important from the firefighting perspective. The objective function of the CP-SAT model then maximizes the total reward accumulated over all time steps and cells, so that under the coverage constraints, the planned search paths are biased toward spatial and temporal regions with higher fire criticality.

Constructing the Reward Matrix $r(t, m)$ for CP-SAT.

- Read the CSV file and populate a lookup table for efficient access.
- To incentivize earlier actions, a time discount factor is applied for each time step t .
- Construct the Reward Matrix $r(t, m)$.

◆ If the node is a wall: $r(t, m) = 0$

◆ Otherwise:

$$r(t, m) = \lfloor 100 \cdot \text{score dict}(t, i, j) \cdot \text{time weight}(t) \rfloor. \quad (30)$$

This process yields a 3D reward matrix $r(t, m)$ that incorporates a reward for early visits.

Pre-compute the Reachable Sets for Movement and the Vision Sets.

- **Reachable Sets.** For each grid cell, a Breadth-First Search (BFS) is performed to explore all reachable nodes within a distance threshold $s(t, p)$. The search is constrained by the map's layout, ensuring that the traversal paths do not pass through walls.
- **Vision Sets.** The vision set for a point p (i.e., the set of grid cells visible from p) is computed using a method analogous to BFS. However, instead of a step limit, this search is constrained by a visibility distance threshold, $c(t, p)$.
- As a further pre-processing step for the CP-SAT model, we **compute the set** $\text{watchers}(t, m)$. This set is formally defined as

$$\text{watchers}(t, m) = \{ p \mid m \in \text{vision}(t, p) \}. \quad (31)$$

In other words, $\text{watchers}(t, m)$ represents the set of all locations p from which cell m is visible at time t .

6.3.2 Constraints

- **Unique-location Constraint.** Each firefighter can be at exactly one node at any time step:

$$\sum_m w_{n,t,m} = 1, \quad \forall n, t. \quad (32)$$

- **Initial-position constraint.** For example, all firefighters start from the same cell with index start id:

$$w_{n,0,\text{start id}} = 1, \quad \forall n. \quad (33)$$

Alternatively, each firefighter can be assigned its own initial node.

- **Movement connectivity constraint (using $\text{reachable}(t, p)$).** If a firefighter n is at a node p at a time step t , then at time $t + 1$ he or she must appear in the set of nodes reachable from p :

$$\text{if } w_{n,t,p} = 1, \quad \sum_{m \in \text{reachable}(t,p)} w_{n,t+1,m} \geq 1, \quad \forall n, t < T - 1, p. \quad (34)$$

Our implementation already enforces this forward connectivity. To avoid any teleportation, one may additionally state a reverse constraint: any node occupied at time $t + 1$ must come from one of its neighbors at time t .

- **Logical relationship between y and w .**

$$y_{t,m} \geq w_{n,t,m}, \quad \forall n, t, m, \quad (35)$$

$$y_{t,m} \leq \sum_n w_{n,t,m}, \quad \forall t, m. \quad (36)$$

In this way, $y_{t,m}$ implements the OR "is there anyone at the node m at time t ".

➤ **Definition of Variable b .**

$$b_{t,m} \in \{0, 1\} \quad (37)$$

$b_{t,m} = 1$ when a node m is searched up to time step t . Equivalently, $\exists i \in [0, t]$, s.t., $v_{i,m} = 1$

➤ **Definition of the visibility variable v (using watchers(t, m)).** For every pair (t, m) :

- ◆ If the set $\text{watchers}(t, m)$ is empty, nobody can see the node m from anywhere, so we enforce $v_{t,m} = 0$.
- ◆ Otherwise:

$$v_{t,m} \geq y_{t,p}, \quad \forall p \in \text{watchers}(t, m), \quad v_{t,m} \leq \sum_{p \in \text{watchers}(t, m)} y_{t,p}. \quad (38)$$

Thus, $v_{t,m}$ means the node m is seen by at least one firefighter standing at some observation node.

➤ **Coverage constraint: each node is seen at least once.** For every node m :

$$\sum_{t=0}^{T-1} v_{t,m} \geq 1. \quad (39)$$

If the requirement is the firefighter must actually walk into the cell to sweep it, replace $v_{t,m}$ with $y_{t,m}$ in the above constraint.

➤ **Additional Constraint.** If a node m is searched up to time step $t - 1$, or it is seen at time step t , then it is searched up to time step t .

$$b_{t+1,m} \geq b_{t,m}, \quad \forall t < T - 1, m \quad (40)$$

$$b_{t,m} \geq v_{t,m}, \quad \forall t, m \quad (41)$$

$$b_{t,m} \leq b_{t-1,m} + v_{t,m}, \quad \forall t \geq 1, m \quad (42)$$

6.3.3 Objective Function

Differentiate the reward function:

$$r'_{t,m} = \begin{cases} r_{t,m} - r_{t+1,m}, & \text{if } t < T - 1. \\ r_{t,m}, & \text{if } t = T - 1. \end{cases} \quad (43)$$

The objective is to maximize:

$$\sum_{t,m} r'_{t,m} \cdot b_{t,m} \quad (44)$$

Interpretation.

- Whichever agent stands on which cell at which time step will obtain the weight of that cell at that moment (with an additional time-discount factor).
- The CP-SAT solver therefore tends to:
 - ◆ Visit rooms that are both highly dangerous and highly important first;
 - ◆ Reach these rooms as early as possible;
 - ◆ While simultaneously satisfying the constraint that every node is seen at least once.

6.3.4 Real-life Applications: The Different Rooms

Office rooms are assigned a risk score; the higher the score, the greater the danger, as shown below.

$$\left\{ \begin{array}{l} 1, \text{ Light Hazard.} \\ 2, \text{ Ordinary Hazard Gr.1.} \\ 3, \text{ Ordinary Hazard Gr.2.} \\ 4, \text{ Extra Hazard Gr.1.} \\ 5, \text{ Extra Hazard Gr.2} \end{array} \right. \quad (45)$$

In our model, we assign each of these five categories a hazard coefficient $\in \{1, 2, 3, 4, 5\}$, this is then multiplied by the fire intensity provided by the Cellular Automaton, to determine the instantaneous risk of a room. Thus, we have

$$\text{Instantaneous risk} = \text{Hazard coefficient} \times \text{Fire intensity.} \quad (46)$$

6.4 Model Results

The results of the prediction are as shown below:



Figure 11: Result for School Lab Floor 1

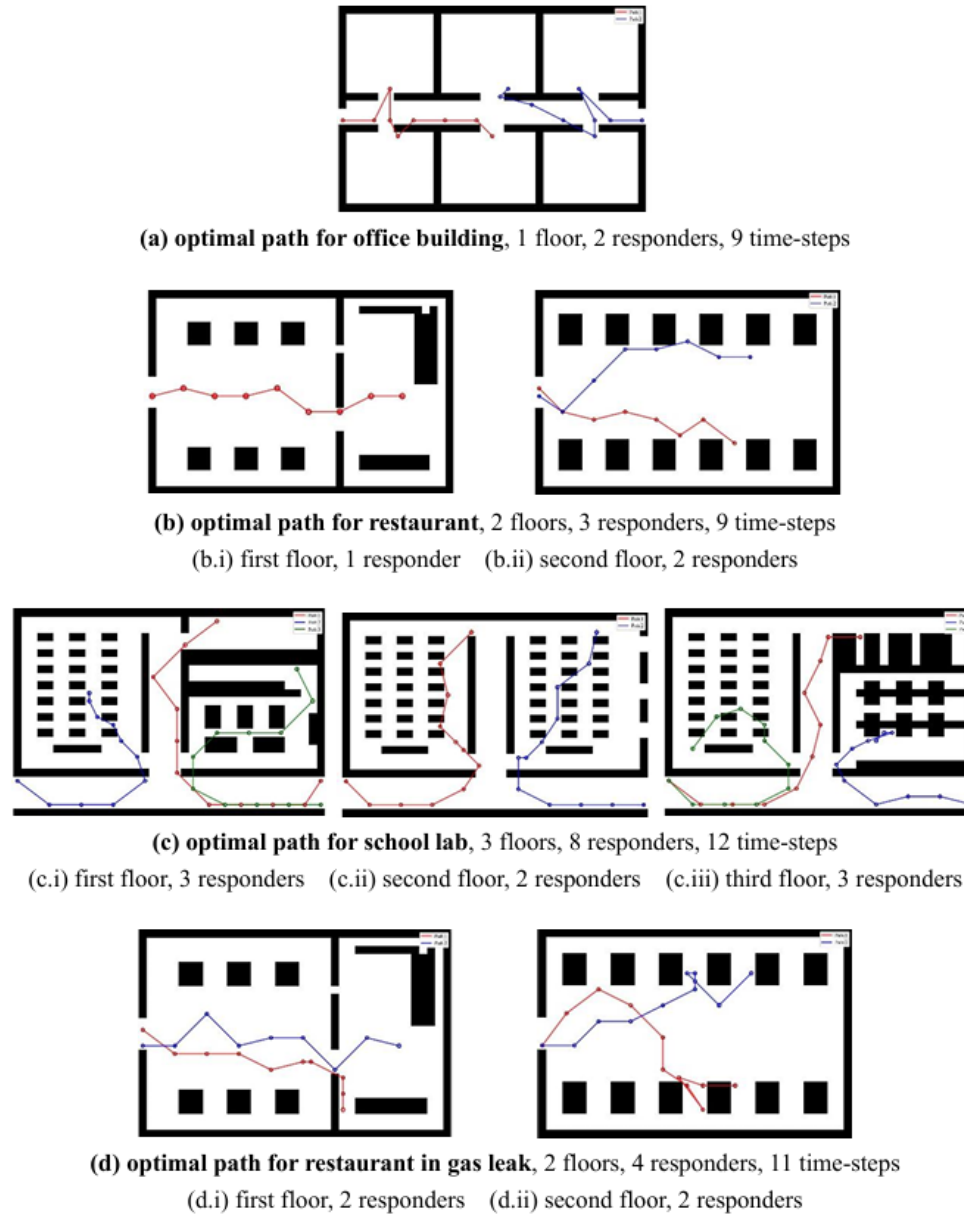


Figure 12: Optimum Path for Different Cases

7 Sensitivity Analysis

We conduct a sensitivity analysis to test the robustness of our model and to identify which key parameters, such as the number of responders or fire spread rate, have the most significant impact on sweep efficiency. This validates the reliability of our conclusions and allows us to provide the most valuable and actionable recommendations to the emergency planning committee.

7.1 Theoretical Basis of FAST

The Fourier Amplitude Sensitivity Test (FAST) is a variance-based global sensitivity method. For a model $Y = f(X_1, \dots, X_k)$, the goal is to quantify how much of the output variance $\text{Var}(Y)$ is contributed by each input X_i . The first-order sensitivity index of X_i is defined as $S_i = V_i / \text{Var}(Y)$, where V_i is the variance due to X_i alone. FAST estimates these indices by mapping all inputs onto a single scalar sampling variable $s \in [-\pi, \pi]$: each input is written as a periodic function $X_i(s)$ with a distinct frequency ω_i . Running the model along this one-dimensional trajectory produces a sequence $Y(s_j)$ whose Fourier spectrum encodes the influence of each input. The total variance $\text{Var}(Y)$ is recovered from the overall spectral energy, while the contribution V_i is obtained by summing the spectral energy at the frequency band associated with ω_i and its harmonics. In this way, FAST converts the high-dimensional variance decomposition problem into a simple Fourier analysis of $Y(s)$.

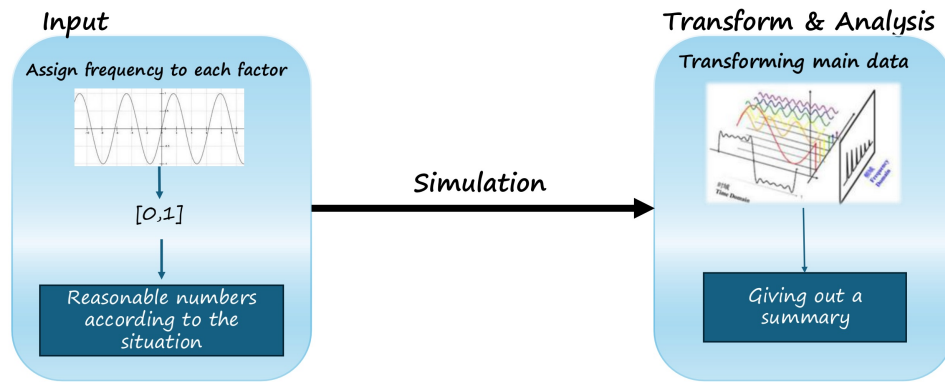


Figure 13: Principle of FAST

7.2 Application in our Model

In our case, applying FAST to these samples yields first-order indices S_i that quantify how much each parameter contributes to the variability of Y . In practice, we focus calibration and refinement on parameters with large S_i , while fixing low-sensitivity parameters at nominal values, which reduces the overall prediction error due to parameter uncertainty without changing the structure of the underlying CA—CP-SAT model.

8 Strengths and weaknesses

What sets our approach apart within the field of modeling is the innovation of our hybrid approach that intertwines an environmentally realistic simulation with mathematical optimization. This makes it so that the strategies we create for sweeping not only operate at a fast pace but do so with an awareness of the danger involved.

The major drawbacks of the proposed model are that the response of the system depends on its initial set of parameters, which should be calibrated on real-life scenarios to maximize the system's accuracy. Additionally, the fact that it's an open-loop system implies that the system cannot adapt to changes that might result due to the activity of the responders.

9 Conclusion

We predicted the optimal result when the agent number is 1, 2, and 3 in all three types of building. As shown by the result, the lab require more time and responders to achieve an optimal sweep strategy due to its complex structure and dangerous chemical materials. In contrast, office buildings and restaurants require fewer responders yet still result in lesser sweep time, which is align with our prediction since office buildings and restaurants have simpler structures and fewer obstacles.

In the case of a gas leak, the number of responders required for optimal sweep routes is larger than that of the fire emergency. This finding also meets our prediction, given that occupants encounter a delayed awareness in gas leak, resulting in a slower evacuation process.

Letter to LEPC

Dear Local Emergency Planning Committee,

We carried out a study on how to effectively plan sweeps and assign responders in emergencies, such as fires or gas leaks. We used a Cellular Automaton (CA) to predict the propagation of fire, smoke and evacuating occupants, and a CP-SAT-based MILP for the optimal number, starting position and sweeping path of the responders.

Our map used in the study usually consists of a main corridor and several branches. We observed that in nearly all cases, one responder rushes toward the branch where fire is most intensive (usually where combustion begins), while others quickly scans other branches. In the optimal solution, every responder focus on their own part of building, without retreating out of a branch or colliding with other responders. For example, a corridor with one classroom on each side requires two responder, while only one is enough for a large, though open space.

When planning sweeps and assigning responders, the following policies should be adapted in order to ensure both redundancy and safety:

- **Every dead end should be assigned to a different responder.** Our study showed that retreating out of a branch is significantly wasteful of time.
- **Avoid overlapping paths.** If the path of multiple responder overlaps, an opportunity of searching additional area is missed.

Our simulation of occupant evacuation presented the importance of building structure and occupant speed. Therefore, we need to formulate a different strategy for different types of building types.

- **Office Buildings.** Responders should first clear the room on fire and rooms nearby. Searching of distant rooms need not be too haste, as few combustible leads to a low propagation rate of fire.
- **Public assemblies.** In our simulation of restaurant, occupants are easily stuck between obstacles or go in the wrong direction due to lack of conduct and coordination. Thus, it's important to effectively provide authoritative instruction and produce a steady flow of people.
- **Schools.** Schools contain not only high density of young occupants, but also complicated structure of desks and chairs. Responders should carefully verify every corners where a small children might seek refuge.

In conclusion, our evaluation result provides detailed evacuation plan and sweeping strategy for various types of buildings. These recommendations gives a concrete framework for training, drills, and ultimately, for saving lives when seconds count.

We sincerely hope our recommendations can be adopted.

Best Regards,

Team #16588

References

- [1] Cheng, H., & Hadjisophocleous, G. V. (2011). Dynamic modeling of fire spread in building. *Fire Safety Journal*, 46(4), 211–224. <https://doi.org/10.1016/j.firesaf.2011.02.003>
- [2] Hadjisophocleous, G. V., & Benichou, N. (1999). Performance criteria used in fire safety design. *Automation in Construction*, 8(4), 489–501. [https://doi.org/10.1016/S0926-5805\(98\)00096-X](https://doi.org/10.1016/S0926-5805(98)00096-X)
- [3] Luh, P. B., Wilkie, C. T., Chang, S.-C., Kerry L. Marsh, & Olderman, N. (2012). Modeling and optimization of building emergency evacuation considering blocking effects on crowd movement. *IEEE Transactions on Automation Science and Engineering*, 9(4), 687–700. <https://doi.org/10.1109/TASE.2012.2200039>
- [4] Ma, G., & Wu, Z. (2020). BIM-based building fire emergency management: Combining building users' behavior decisions. *Automation in Construction*, 109, 102975. <https://doi.org/10.1016/j.autcon.2019.102975>
- [5] Shi, J., Ren, A., & Chen, C. (2009). Agent-based evacuation model of large public buildings under fire conditions. *Automation in Construction*, 18(3), 338–347. <https://doi.org/10.1016/j.autcon.2008.09.009>
- [6] Simulation of fire spread between residential buildings regarding safe separation distance. (2017). *Tehnicki Vjesnik - Technical Gazette*, 24(4). <https://doi.org/10.17559/TV-20150923233514>
- [7] Zheng, Y., Jia, B., Li, X.-G., & Zhu, N. (2011). Evacuation dynamics with fire spreading based on cellular automaton. *Physica A: Statistical Mechanics and Its Applications*, 390(18–19), 3147–3156. <https://doi.org/10.1016/j.physa.2011.04.011>
- [8] Alvelos, F. (2018). Mixed integer programming models for fire fighting. In O. Gervasi, B. Murgante, S. Misra, E. Stankova, C. M. Torre, A. M. A. C. Rocha, D. Taniar, B. O. Apduhan, E. Tarantino, & Y. Ryu (Eds.), *Computational Science and Its Applications – ICCSA 2018* (Vol. 10961, pp. 637–652). Springer International Publishing. https://doi.org/10.1007/978-3-319-95165-2_45
- [9] Curiac, D. I., Baniias, O., Volosencu, C., & Pescaru, D. (n.d.). *Cellular automata based simulation for smoke and fire spreading in large buildings*.
- [10] De Ris, J. N. (1969). Spread of a laminar diffusion flame. *Symposium (International) on Combustion*, 12(1), 241–252. [https://doi.org/10.1016/S0082-0784\(69\)80407-8](https://doi.org/10.1016/S0082-0784(69)80407-8)
- [11] Huang, H., & Li, T. (2020). Cellular automata simulation model for emergency evacuation planning. *Construction Research Congress 2020*, 468–477. <https://doi.org/10.1061/9780784482865.050>
- [12] Liu, C., Mao, Z., & Fu, Z. (2016). Emergency evacuation model and algorithm in the building with several exits. *Procedia Engineering*, 135, 12–18. <https://doi.org/10.1016/j.proeng.2016.01.072>
- [13] ShanShan Ma. (2023). Simulation study of pedestrian evacuation in museums based on potential energy field. *Open Journal of Transportation Technologies*, 12(4), 245–259. <https://doi.org/10.12677/OJTT.2023.124028>